

U.S. SEMICONDUCTOR: TRADE & LABOR ANALYSIS

Dashboard 1 — U.S. Semiconductor Imports Overview (CIF by Country, 2025)

Overview

This document describes the methodology, components, and key findings of Dashboard 1 — the U.S. Semiconductor Imports Overview (CIF by Country, 2025). The dashboard was developed to provide a comprehensive, data-driven view of U.S. semiconductor import flows in 2025, with a focus on Cost, Insurance, and Freight (CIF) values by exporting country.

The analysis draws on official UN Comtrade Plus trade statistics and was validated independently across Python, SQL, and Excel before final visualization in Power BI. The dashboard is structured to reveal both aggregate market dynamics and country-level supply chain dependencies, with dual views — with and without Malaysia — to isolate the full scale of concentration risk.

Project Context

Attribute	Value
Dataset Source	UN Comtrade Plus (official trade statistics)
Scope	U.S. semiconductor imports in 2025 — CIF (Cost, Insurance, Freight) values
Total CIF Value	\$11.58bn (with Malaysia) / \$1.96bn (without Malaysia)
Dataset Rows	288 (with Malaysia) / 244 (without Malaysia)
FOB Reference	\$82.11M (Brazil & Mexico)

Tools Used

Tool	Purpose
Python	Exploratory Data Analysis (EDA), statistical validation, visualizations
SQL	Independent validation, raw number extraction, precise market share calculations
Excel	Pivot tables, CIF vs. FOB gap analysis, validation layer
Power BI	Final interactive dashboard for storytelling and stakeholder engagement

Dashboard Components

Component	Description
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KPI Cards	Total CIF imports, dataset rows, FOB values, CIF–FOB gap
Line Chart	Monthly import values by exporter country (Jan–Dec 2025)
Donut Chart	Market share distribution by country
Bar Chart	Transport value by exporter country
Drill-through Pages	Country-level profiles with transport mode breakdowns

Methodology

Data Cleaning

Zero-value transactions were removed from the raw dataset to ensure that reported CIF totals reflect only active trade flows. Country codes were standardized to resolve discrepancies between UN Comtrade naming conventions and Power BI display labels. Schema validation was performed in SQL to confirm column integrity and data type consistency prior to analysis.

Cross-Validation

All key metrics — total CIF, country-level shares, and CIF–FOB gap calculations — were independently calculated in Python, SQL, and Excel. Results matched to the penny across all three tools, confirming the integrity of the underlying data and the accuracy of the dashboard figures.

Visualization Choices

- Line charts: Used for temporal trend analysis, showing monthly import values by exporter country across January–December 2025.
- Donut charts: Used for market share clarity, highlighting proportional concentration among exporting countries.
- Bar charts: Used for transport economics, comparing transport value by exporter country.
- Conditional formatting in Power BI: Applied to highlight Malaysia's dominant position relative to emerging market suppliers.

Dual View Design

The dashboard was constructed with two parallel views — one including Malaysia and one excluding it. This design choice was deliberate: Malaysia's ~60.5% share compresses all other country values to near-zero on shared axes, making it impossible to analyze the remaining supplier landscape without filtering it out. The dual view preserves the full picture while enabling actionable insight into the non-Malaysia supply base.

Executive Summary: Pivotal Insights

Country	CIF Share	Ex-Malaysia Share	Strategic Profile
Malaysia	~60.5%	~83.1% transport share	Dominant supplier — structural dependency risk
Germany	~10.2%	~32% ex-Malaysia	Premium chips, air freight-heavy, limited scale
Japan	~7.8%	~25% ex-Malaysia	Consistent mid-tier supplier, steady partner
Brazil	~2.8%	~9% ex-Malaysia	Balanced transport strategy, growing consistency

India	~2.5%	~8% ex-Malaysia	Cost leader, high shipment volume
Israel	~1.6%	~5% ex-Malaysia	Specialist, high-value shipments
Hong Kong SAR	~4.3%	~24.79% ex-Malaysia	China routing gateway — geopolitical complexity
Canada / Mexico	<0.1%	Negligible	Virtually absent — USMCA proximity irrelevant here

Detailed Findings

1. Malaysia's Overwhelming Dominance

- Market Share: ~60.5% of all U.S. semiconductor imports in 2025.
- Transport Share: ~83.1% of total transport value, reflecting the scale of freight associated with its shipment volumes.
- Implication: Structural dependency on Malaysia's supply chain. Reliable under normal conditions but represents a single point of failure for U.S. semiconductor access.

2. Germany as the Western Alternative

- Market Share: ~10.2% overall; rises to ~32% when Malaysia is excluded from the denominator.
- Profile: Premium, specialized chips with a heavy reliance on air freight, consistent with high-value, time-sensitive shipments.
- Takeaway: Germany is the strongest Western ally and fallback supplier, but capacity constraints limit its ability to absorb Malaysia's share at scale.

3. Japan's Stability

- Market Share: ~7.8% overall; ~25% excluding Malaysia.
- Profile: Consistent, predictable, moderate volume — a reliable mid-tier supplier.
- Takeaway: Japan offers steady, dependable supply and is a strong candidate for expanded partnership in a diversification strategy.

4. Emerging Trio: Brazil, India, Israel

- Brazil: Balanced transport strategy and growing shipment consistency. ~9% share excluding Malaysia.
- India: Cost leader with high shipment volume. ~8% share excluding Malaysia.
- Israel: Specialist supplier of high-value, low-volume shipments. ~5% share excluding Malaysia.
- Takeaway: Each country fills a distinct niche — volume, cost efficiency, and technical specialization. Together, they form a natural diversification portfolio.

5. Hong Kong as China's Gateway

- Insight: Hong Kong exports more semiconductors to the U.S. than Shanghai on a direct-reporting basis.
- Implication: Hong Kong's elevated role reflects a routing strategy designed to reduce direct China–U.S. trade scrutiny. This pattern warrants ongoing monitoring as U.S.–China trade policy evolves.

6. Canada & Mexico Absence

- Finding: Canada and Mexico are virtually absent from U.S. semiconductor import flows in 2025.
- Implication: Geographic proximity and USMCA trade agreements do not translate into semiconductor trade relevance. North American nearshoring for semiconductors remains underdeveloped.

Clear Takeaway

The U.S. semiconductor import landscape in 2025 is highly concentrated, with Malaysia controlling the majority share. Germany and Japan provide reliable alternatives with established supply relationships, while Brazil, India, and Israel represent strategic growth opportunities for supply chain diversification. Hong Kong's disproportionate role highlights the geopolitical complexity embedded in current trade routing, and Canada and Mexico's absence challenges assumptions about the relevance of regional proximity in semiconductor trade.

Strategic Recommendations

- Mitigate Malaysia dependency through proactive supply chain diversification across multiple geographies and transport modes.
- Strengthen partnerships with Germany and Japan as the most reliable high-capacity alternatives to Malaysia.
- Invest early in Brazil, India, and Israel to build long-term supply resilience across volume, cost, and specialization dimensions.
- Monitor Hong Kong's role in China–U.S. trade dynamics as policy scrutiny on indirect routing intensifies.
- Assess USMCA-aligned semiconductor development opportunities in Canada and Mexico as a longer-term nearshoring option.

Data Source

[UN Comtrade Plus — Official International Trade Statistics Database](#)